

Static Analysis and Verification

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Some introductory notes

- ❖ These slides are intended for the course of **Static Analysis and Verification** at FIT BUT.
- ❖ Students are assumed to have a basic knowledge of (and, perhaps even more importantly, *to like* or, at least, *not to be afraid of*) **theories of automata, graphs, logics, algebra, and modelling**.
- ❖ Students are expected to themselves **actively search and study** further information available in recommended textbooks and on the Internet.
- ❖ Students are also strongly advised to **experiment with available tools** for static (formal) analysis and verification in order to get deeper understanding of the capabilities and limits of the presented techniques.
- ❖ Note that many of the presented subjects are **relatively new, still under a very active research**, they are **rapidly developing**, and also the **terminology is not always completely uniform and stable**.

References, Inspiration

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Basic Terminology

Analysis and Verification

❖ **Verification** is a process of checking whether a given **system** (a real system or a model) satisfies a given **correctness specification** (property).

- For example, given a concurrent program, we ask whether it is deadlock-free.
- A verification method provides (ideally) **yes/no answers** having the sense that the system is/is not correct wrt. the given specification.
- The yes/no answer may be complemented by some **diagnostic information** (e.g., when a deadlock is possible, a run leading to some deadlock situation is provided, or a certificate/proof that no deadlock is possible is produced).

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❖ **Analysis** finds answers to **more general questions** about a given system—either questions which are not Boolean, or even if they are, they do not necessarily directly speak about correctness of the system.

- For example, given a concurrent system, we ask whether there is some ordering over the shared resources in which they always get locked.
- Answers provided by analysis may serve as **input for further reasoning** leading to the verification of the system. However, they can also be used for completely **different purposes**, e.g., for optimisation, synthesis/code generation, etc.

Formal Methods

❖ Verification and analysis methods include:

- “bug hunting”: simulation, testing and dynamic analysis, some forms of static analysis, bounded model checking, ...
- formal analysis and verification: model checking, some forms of static analysis, theorem proving, ...

❖ Formal methods are naturally based on formal, mathematical roots—*this does not imply that they are necessarily manual and not usable by ordinary users (!)*.

❖ Unlike other approaches, formal verification is (at least potentially) capable of proving correctness of a given system wrt. a given specification—and not just disprove the property based on some observed behaviour.

❖ Similarly, formal analysis is capable of providing answers universally (conservatively) covering all possible behaviours of a given system (not just giving existential answers—though perhaps extrapolated—based on some observed behaviours).

The Ideal of Formal Verification

- ❖ **Full automation**: no human help needed.
- ❖ **Soundness**: if a verification method claims that a system is correct wrt. a given specification, it is indeed correct.
- ❖ **Completeness**: if a verification method claims that a system is not correct, there is indeed an error in the system—i.e., no **false alarms** (false positives).

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 - Likewise, some sources tolerate that an **indefinite answer** “don’t know” may appear, some exclude it.
- ❖ **Hard to achieve in general**:
 - the **state explosion problem**—i.e., an exponential growth of the number of reachable states wrt. the source description of a finite-state system:
 - `int n`; can have 2^{32} (or 2^{64} , 2^{128} , ...) possible values,
 - m such variables: $2^{m \cdot 32}$ (or $2^{m \cdot 64}$, $2^{m \cdot 128}$, ...) possible values,
 - n concurrent processes each with m states can generate m^n states,
 - **undecidability**: it suffices to have two *unbounded* integer variables, operations `++` and `--`, and branching according to equality with zero.

Relaxing the Ideal

- ❖ A verification method needs **not guarantee termination** and/or can produce **false alarms**.
- ❖ Alternatively, a method is allowed to stop with a “**don't know**” answer and/or becomes **not fully automated** (i.e., some human—or, possibly, AI—help is required).
- ❖ Sometimes, even soundness is sacrificed to efficiency leading to an **bug hunting method** with formal verification roots.
 - For example, possible errors are **ranked** according to chances they are real, and only warnings of at least some rank are shown.

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- ❖ Sometimes, even soundness is sacrificed to efficiency leading to an **error detection method** with formal verification roots.
 - For example, possible errors are **ranked** according to chances they are real, and only warnings of at least some rank are shown.
- ❖ Note that even **if full formal verification of a system fails, it may still be useful as it can find some errors** in the mean time.
- ❖ The **errors found by formal methods can differ** from those found by other methods due to other principles on which the methods work.
 - Hence, it is often a good idea **to use as many different approaches as possible**.

Soundness

- ❖ Introduced in [Livshits *et al*, CACM'15].
- ❖ Soundness not necessary for most modern analysis applications:
 - many clients can tolerate unsoundness.
- ❖ A **soundy analysis** aims to be **as sound as possible** without excessively compromising precision and/or scalability and to **give up soundness in a controlled way**.
- ❖ One may **distinguish**
 - analyses that **are soundy**, i.e., mostly sound, with specific, well-identified unsound choices (e.g., reflection when analysing Java code),
 - analyses that **do not concern themselves with soundness at all**.

Systems and Properties To Be Checked

Systems To Be Checked

- ❖ Systems to be verified/analysed can be classified in many different ways:
 - **Real systems** (SW in common programming languages, HW in VHDL, Verilog, ...) and **models** (based on process algebras, Petri nets, specialised modelling and specification languages like SMV, Promela, Simulink, EARS, CLEAR, OPM ...).
 - Note: we may distinguish **high-level specification of systems** and **specification of properties** (next slide), but the border is blurry.
 - **Finite-state** and **infinite-state systems**—dealing with parameterisation, unbounded recursion (stacks), queues (channels), counters (integer variables), dynamic instantiation, continuous phenomena (time in **real-time systems**, ...), ...
 - **Sequential** and **concurrent systems**—in the latter case, **asynchronous or synchronous**, with **interleaving or true concurrency**, ...

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 - **Sequential** and **concurrent systems**—in the latter case, **asynchronous or synchronous**, with **interleaving or true concurrency**, ...
 - **Control-intensive** and **data-intensive systems**.
 - **Transformational** and **reactive systems**:
 - **Transformational systems** are designed to transform a certain input to a certain output (compilers, database queries, ...).
 - **Reactive systems** typically transform an infinite sequence of inputs to an infinite sequence of outputs (control systems, operating systems and their sub-systems, ...).

Properties To Be Checked

❖ Properties to be verified/analysed can again be classified in various ways:

- The **specification language**:

- the property may be fixed for a given method/tool/plugin,
- a choice out of a list,
- various kinds of **labels/inscriptions** (assertions, end-state labels, progress labels, invariants, ...),
- **automata** (FSA, Büchi automata, ...), **temporal logic formulae** (LTL, CTL, CTL*, μ -calculus, ...),
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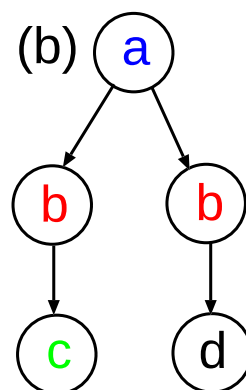
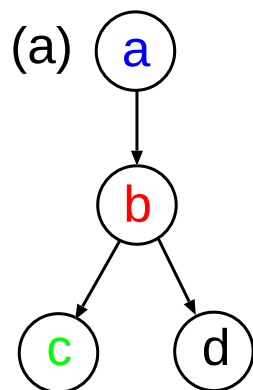
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- The form of **counterexamples** (finite or infinite):
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- A special kind of properties to be checked are various forms of **equality/refinement/simulation** of two system versions.

Notions of Time

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- ❖ **Physical time** allows one to measure how much time passed between two states/events.
- ❖ **Linear time** allows one to speak about particular linear traces of the state space of a given system only.
 - *In all traces, b must happen and be immediately followed by c .* (1)
 - *In all traces, b must happen and be immediately followed by c or d .* (2)
- ❖ **Branching time** allows one to quantify (existentially and universally) over the possible futures of a given state. We view the state space unfolded into an infinite tree.
 - *b must happen, and immediately after, c may happen and d may happen.* (3)



	(a)	(b)
(1)	No	No
(2)	Yes	Yes
(3)	Yes	No

Linear Time Safety and Liveness

- ❖ **Safety** properties require that **something bad never happens**.
 - Their violation always has a **finite witness**. In other words, if there is a counterexample, there is a finite counterexample.
 - **Examples:**
 - *Processes mutually exclude each other when accessing a shared variable.*
 - *A program never returns a wrong result.* (partial correctness)
 - *No null pointer exception can arise.*
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 - *A signal s_1 never arrives before a signal s_2 .*
- ❖ **Liveness** properties require that **something good eventually happens**.
 - Their violation can only have an **infinite witness**, or a **finite but complete witness**, i.e., a witness that cannot be extended any more.
 - **Examples:**
 - *A program terminates for any input.* (total correctness)
 - *A process can never starve.*
 - *After a signal s_1 is received, a signal s_2 is eventually sent.*
 - Liveness is typically **harder to verify** than safety.
- ❖ **General properties** can mix (conjoin) safety and liveness.

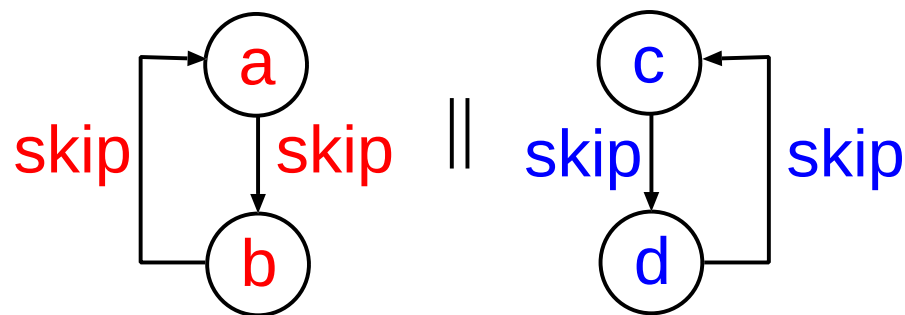
Branching Time Safety and Liveness

- ❖ The situation is **more complicated** for branching time: a computation tree can be infinite and incomplete at the same time!
- ❖ One can distinguish [Manolios, Trefler – LICS 2001]:
 - **Universally safe properties**: linear time safety over all computations.
 - **Existentially safe properties**: guarantee at least one safe computation.
 - **Universally live properties**: linear time liveness properties over all computations.
 - **Existentially live properties**: linear time liveness for at least one computation.
- ❖ **General properties** are **intersections** of some of the above kinds of properties.

Fairness

❖ In order to **exclude practically infeasible, trivial counterexamples to liveness properties**, one usually has to apply some **fairness assumptions**.

- For example, assume that we have two concurrent, infinitely looping processes, not requesting anything from their environment.



- We ask whether each of them will always eventually take a step.
- We somehow naturally expect the answer to be **yes**, but this is based on that we **implicitly assume** that the processes are scheduled by **some fair scheduler**.
- However, if we do not build the scheduler into the system, the answer will be **no**.
- Alternatively, we can perform the verification under a **suitable fairness assumption**—either expressed as a part of the property being checked or hard-wired into the verification algorithm.

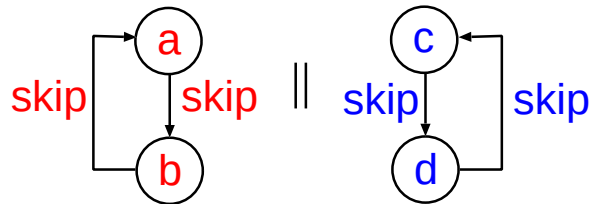
Fairness

❖ Fairness **limits sources of non-determinism** in a system (e.g., in the scheduling or in reading random data from the environment).

❖ Two **most common kinds of fairness**:

- **Weak fairness:**

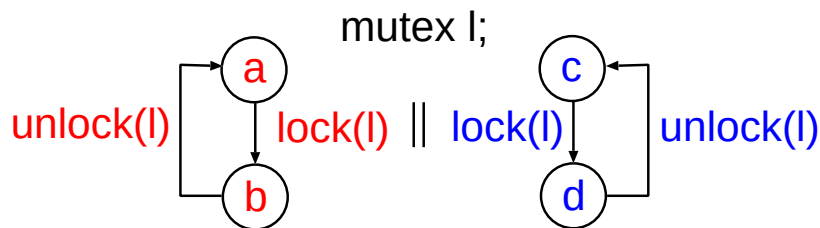
- an action that is eventually always enabled must always eventually be taken,



- enough for non-synchronizing processes.

- **Strong fairness:**

- an action that is always eventually enabled must always eventually be taken,
- needed when dealing with fair resource allocation,



- more complicated to handle.

Different Approaches to Formal Verification and Analysis

Model Checking

[Clarke, Emerson 81], [Quielle, Sifakis 81]

❖ An algorithmic approach of checking whether a given system satisfies a given property through a systematic exploration of the **state space** of the system.

- The system may be a **real system** or a **model**.
 - Sometimes the use of models is stressed.
- Usually, **finite-state systems** are considered.
 - Dealing with infinite-state systems is less common.
- Properties classically specified using **temporal logics** (LTL, CTL, CTL*, ...); other forms of specification are also possible.

Model Checking

❖ **Advantages** include:

- a (relatively) high degree of automation—fully automated up to the need of modelling the system, its parts, and/or its environment,
- (relatively) easy to use,
- quite general as for the systems and their properties that can be checked,
- provides counterexamples.

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❖ Disadvantages:

- the **state explosion problem** (and dealing with infinite-state spaces),
- a need to model the environment of the checked (sub-)system.

Model Checking

❖ Dealing with the **state explosion problem**—just a brief overview, we will get back to (at least some of) the mentioned approaches later on:

- **Efficient storage** of state spaces (hierarchical storage of states, BDDs, ...).
- State space **reductions** (symmetries, partial-order reduction, ...).
- **Abstraction**, counterexample-guided abstraction refinement (CEGAR).
- **Compositional methods**, assume-guarantee reasoning.
- [**Bounded model checking**: exploring the state space up to some bound only (sacrificing soundness), may leverage advances in SAT/SMT solving.]

❖ Especially suitable for **reactive, concurrent, control-intensive systems**.

❖ Supported by many **tools**, including industrial-strength tools:

- Spin, Divine, Blast, CPAchecker, CBMC, JBMC, Ultimate Automizer, JPF (NASA), NuXMV, ABC, Incisive Formal Analysis (Cadence), Questa (Siemens), VC Formal (Synopsys), Uppaal, Prism, ...

Static Analysis

- ❖ Collects some information about the behaviour of a system from its **source code** without actually executing it under its original semantics.
- ❖ This description is very general, it can include even model checking and theorem proving (at least in some of their forms), which is sometimes (but not usually) done.
- ❖ Static analyses **range** from simple syntactic checks to iterative fixpoint computations over an abstraction of the examined system (e.g., into a form of equations that need to be solved).
- ❖ Different **forms of static analysis**:
 - bug pattern (anti-pattern) searching,
 - dataflow analysis,
 - constraint-based analysis,
 - (extended) **type analysis** (type and effect systems),
 - abstract interpretation,
 - **symbolic execution**, ..., mixtures of the above.

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❖ **Disadvantages**:

- can produce many **false alarms**
 - construction of more and more precise analyses leads to similar efficiency problems like in model checking,
 - often resolved by accepting unsoundness,
- various analyses are often **specialised** just for a certain specific task,
- can have in principle **restricted applicability** for some properties of interest (e.g., not everything can be simply expressed by a bug pattern).

Static Analysis

- ❖ There exist **many tools** for static analysis too:
 - FindBugs/SpotBugs, Synopsys/Coverity, Klocwork, CodeSonar (GramaTech), PolySpace, Code Analysis in VisualStudio, AbsInt/Astrée, Clang Static Analyser, gcc static analysis, Cppcheck, cppclean, Sparse, Meta/Facebook Infer, Facebook SPARTA, Frama-C, Predator, KLEE, Loopus, Cost, Symbiotic, ...

Theorem Proving

- ❖ Deductive verification often similar to the classical mathematical way of proving theorems starting with axioms and inferring further facts using rules of correct inference.
- ❖ Advantages and disadvantages:
 - Very general.
 - Usually semi-automated, requires a significant manual effort (large space for AI!).
 - Problems with diagnostic information for incorrect systems.
- ❖ There exist many interactive theorem provers including:
 - PVS, Coq, Hol, Isabelle, ACL2, Forte, ...
- ❖ Used, e.g., by Intel, AMD to verify complex designs of various arithmetic units (sometimes combined with model checking). Ongoing attempts to arrive to formally verified operating system kernels (seL4), compilers (CompCert), ...

Theorem Proving

❖ Recently there has been a lot of progress on fully automated **decision procedures** for various decidable theories:

- **propositional logic** (SAT solving),
- various **first-order theories** (SMT solving)
 - uninterpreted functions,
 - linear integer/real arithmetic,
 - theories of arrays, bitvectors, strings, ...
- WS1S, WS2S, ...

❖ There exist many **tools** implementing various decision procedures:

- **SAT solvers**: Kissat, CaDiCaL, MapleSAT, glucose, ...
- **SMT solvers**: Z3, CVC5, MathSat, Yices2, Boolector, SMTinterpol, ...
- WS1S/WS2S: Mona, Gaston, ...

❖ There exist **fully automated theorem provers** for **undecidable logics** as well: e.g. Vampire for predicate logic (no guarantee of termination).

Theorem Proving

- ❖ Decision procedures often serve as a **back-end** for other verification approaches, e.g.:
 - **Predicate abstraction** for model checking.
 - Deductive verification based on discharging **verification conditions**.
 - Code is **annotated** by loop invariants, pre-/post-conditions of the program and procedures, and/or various assumptions.
 - Annotations can be **(partially) obtained automatically**—e.g., from analysing runs of the system and generalising them, ruling out candidate constraints describing loops through test runs, **using LLMs**, etc.
 - Annotations allow verification to concentrate on loop-free fragments.
 - **Verification conditions**: checking that when one starts under some precondition and executes a loop-free code fragment, the postcondition will be met.
 - Used, e.g., in attempts to verify operating system kernels—PikeOS, Microsoft's Hypervisor (see the tool VCC).
 - **Symbolic execution** – iterating over longer and longer program paths possibly leading to an error, encoding them as formulae, checking satisfiability.
 - Klee, Symbiotic, SPF, Java Ranger, JDart, ...